# L D COLLEGE OF ENGINEERING

COMPUTER ENGINEERING DEPARTMENT

Testing, Analysis and UI For Smart Traffic Signal Controller


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# Testing Methodology

- Test the fuzzy controller with at least 10 different input combinations.
- Two inputs are used:
  - Traffic Density: 0–100%
  - Waiting Time: 0–120 seconds
- The controller calculates:
  - Green Signal Time: 10–90 seconds
- Purpose: Check whether the controller gives suitable green time for different traffic conditions.

# Test Cases

| Test | Density (%) | Waiting (sec) | Green Time |
|---|---|---|---|
| 1 | 10 | 10 | ~20 |
| 2 | 20 | 30 | ~30 |
| 3 | 30 | 50 | ~40 |
| 4 | 40 | 40 | ~43 |
| 5 | 50 | 60 | ~50 |
| 6 | 60 | 60 | ~60 |
| 7 | 65 | 75 | ~68 |
| 8 | 70 | 90 | ~72 |
| 9 | 85 | 100 | ~82 |
| 10 | 95 | 115 | ~88 |

# Sample Testing

- **Input**:

Traffic Density = 65%

Waiting Time = 75 sec

- **Output**:

Green Signal Time = ≈68 sec

- **Result**:

Since both traffic density and waiting time are relatively high, the controller provides a longer green signal duration.

# Testing Observation

- Low density + short waiting → shorter green time
- Medium density + medium waiting → medium green time
- High density + long waiting → longer green time
- Green time changes according to traffic conditions.
- This demonstrates the adaptive behaviour of the fuzzy controller.

# Simulator Testing Results

- **Input values:**
- Traffic Density = 65%
- Waiting Time = 75 sec
- Python code where these values are entered.
- **Fuzzy controller execution :**
- **Code containing:**

  traffic_sim.input['density']

  traffic_sim.input['waiting']

  traffic_sim.compute()
- **Result**:
- Green Signal Time ≈ 68 seconds

# Controller Behaviour

- The fuzzy controller uses Traffic Density and Waiting Time as inputs.
- These inputs are converted into fuzzy values.
- The fuzzy rule base determines the appropriate green-signal category.
- Defuzzification produces the final Green Signal Time.
- As traffic conditions become more demanding, the controller generally provides a longer green duration.
- Example: 65% density + 75 sec waiting → approximately 68 sec green time

# Strengths of the Controller

- **Adaptive**: Green time changes according to traffic conditions.
- **Flexible**: Uses multiple fuzzy rules for different traffic situations.
- **Simple inputs**: Traffic density and waiting time are easy to understand.
- **Easy implementation**: Developed using Python and scikit-fuzzy.
- **Better than fixed timing**: Can adjust signal duration according to current conditions.
- **Expandable**: Additional traffic parameters can be added later.

# Weaknesses of the controller

- Only two input parameters are used in the current controller.
- Membership functions are predefined.
- Fuzzy rules need proper tuning.
- The current simulation does not include all real-world traffic conditions.
- The system does not currently consider pedestrians or emergency vehicles.
- Physical traffic-signal hardware is not connected.

# Scope of Improvement

- Add real-time traffic sensors/CCTV.
- Include emergency vehicle priority.
- Add pedestrian waiting time.
- Integrate AI/ML for traffic prediction.
- Connect multiple signals using IoT.
- Develop a real-time traffic monitoring dashboard.
- Optimize fuzzy membership functions and rules using real traffic data.

# User Interface for Smart Traffic Signal Controller

**Main objectives**:

- Enter traffic density.
- Enter waiting time.
- Run the fuzzy controller.
- Display recommended green time.
- Display traffic condition.
- Show signal status visually.
- Make the system easy to operate.

# Smart Traffic Signal Controller – UI

- **Input Panel**
- Traffic Density: 0–100%
- Waiting Time: 0–120 Sec
- Calculate Signal button
- **Output Panel**
- Traffic Condition: Low / Medium / High
- Recommended Green Time: XX seconds
- Signal Status: RED / YELLOW / GREEN
- **Visual Display**

RED

YELLOW

GREEN

# Working of User Interface

```
User Input
    ↓
Traffic Density + Waiting Time
    ↓
Fuzzification
    ↓
Fuzzy Rule Evaluation
    ↓
Defuzzification
    ↓
Green Time Calculation
    ↓
Traffic Signal Display
```

**Example:**

Density = **85%**
Waiting Time = **90 sec**

↓

Traffic Condition = **High**

↓

Recommended Green Time = **Long**

↓

🟢 **GREEN**

# Features of Developed UI

- Simple and user-friendly interface.
- Input sliders/text boxes.
- Automatic fuzzy calculation.
- Real-time output display.
- Traffic condition indicator.
- Traffic signal visualization.
- Recommended green-time display.
- Error handling for invalid inputs.
- Can be extended to multiple traffic lanes.

# Final Demonstration

- Sample Output:

| Parameter | Value |
|---|---|
| Traffic Density | 85% |
| Waiting Time | 90 sec |
| Traffic Condition | High |
| Green Time | 60 sec |
| Signal | GREEN |

# Conclusion

- The fuzzy-based Smart Traffic Signal Controller provides adaptive signal timing.
- Testing demonstrates its response to different traffic situations.
- Analysis shows that fuzzy logic is flexible and interpretable.
- The UI makes the controller easy to use and demonstrate.
- The system has potential for integration with real-time traffic sensors and intelligent transportation systems.

# Thank You